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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 8

Title: Creating Marketing Campaign Performance Dashboards

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Problem Statement

You are a Marketing Analyst in a digital marketing agency. Senior management wants to track campaign reach, conversion performance, acquisition cost, and return on investment across channels, audiences, and customer segments. This experiment creates an interactive Marketing Campaign KPI dashboard that helps evaluate the performance of marketing campaigns, using the Marketing Campaign Performance Dataset.

Github Link: <https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Experiment%208%20Marketing%20Campaign%20Dashboard>

Methodology

All KPI cards, charts, and slicers were arranged into a single interactive Marketing Campaign dashboard in Power BI. The dashboard includes:

- Total Campaigns KPI Card
- Avg Conversion Rate KPI Card
- Total Clicks KPI Card
- Avg ROI KPI Card
- Acquisition Cost & ROI by Campaign Type (Combo Chart)
- Campaigns by Customer Segment (Donut Chart)
- Campaigns by Target Audience (Bar Chart)
- Average ROI by Channel (Bar Chart)
- Monthly Conversion Rate (Line Chart)
- Interactive Slicers: Location and Campaign_Type

The dashboard enables management to analyze campaign reach and conversion trends, monitor acquisition cost against ROI, and identify the channels, audiences, and segments driving the strongest performance, supporting data-driven marketing decision-making.

Dashboard Images

Image 1st:

It displays the overall Marketing Campaign dashboard without any filters applied (Location and Campaign_Type slicers set to "All").

- Total Campaigns: 200K and Total Clicks: 110M, indicating a large-scale, multi-channel campaign portfolio.
- Avg Conversion Rate: 8.01% and Avg ROI: 5.00, reflecting a consistent, stable return across the full campaign base.
- The Acquisition Cost & ROI by Campaign Type chart shows all five campaign types (Influencer, Search, Display, Social Media, Email) with near-identical acquisition cost (~₹0.5bn each), while the ROI line dips lowest for Social Media (4.99) before recovering slightly for Email.

- Campaigns by Customer Segment are evenly split across the five segments (~19.9–20.1% each: Foodies, Tech Enthusiasts, Outdoor Adventurers, Health & Wellness, Fashionistas).
- Campaigns by Target Audience are also evenly distributed at ~40K campaigns for each of the five audience groups (Men 18-24, Men 25-34, All Ages, Women 25-34, Women 35-44).
- Average ROI by Channel is uniform (~5.0) across Facebook, Website, Google Ads, and Email, showing no channel is a standout underperformer.
- Monthly Conversion Rate stays in a tight band around 8%, ranging from roughly 7.97% to 8.05% across the year with no strong seasonal pattern.

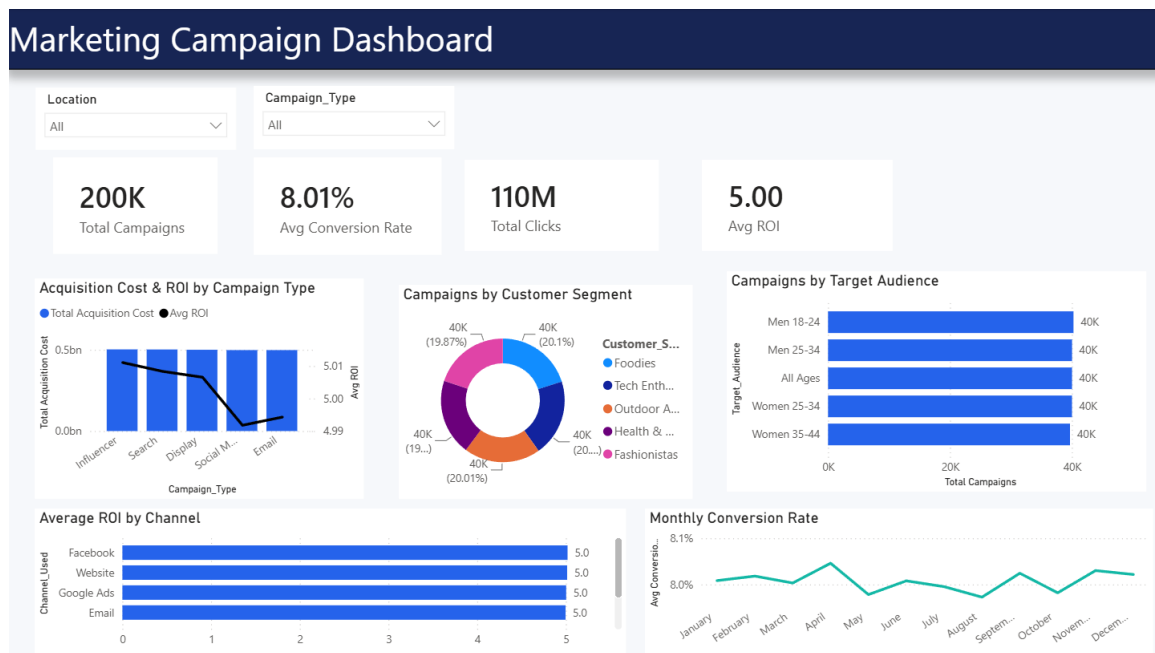


Figure 1: Marketing Campaign Dashboard – Overall View (No Filters Applied)

Image 2nd:

In the dashboard, after selecting the Facebook bar in the Average ROI by Channel chart:

- All visuals are cross-filtered to show data for the Facebook channel only.
- The KPI cards update to Total Campaigns = 33K, Avg Conversion Rate = 8.00%, Total Clicks = 18M, and Avg ROI = 5.02 — slightly above the overall average.
- Campaigns by Customer Segment remain fairly balanced within Facebook (~19.8–20.2% each), and Campaigns by Target Audience stay close to ~6–7K per group.
- This confirms that Facebook-driven campaigns convert marginally better than the portfolio average, making it a channel worth prioritizing for incremental budget.

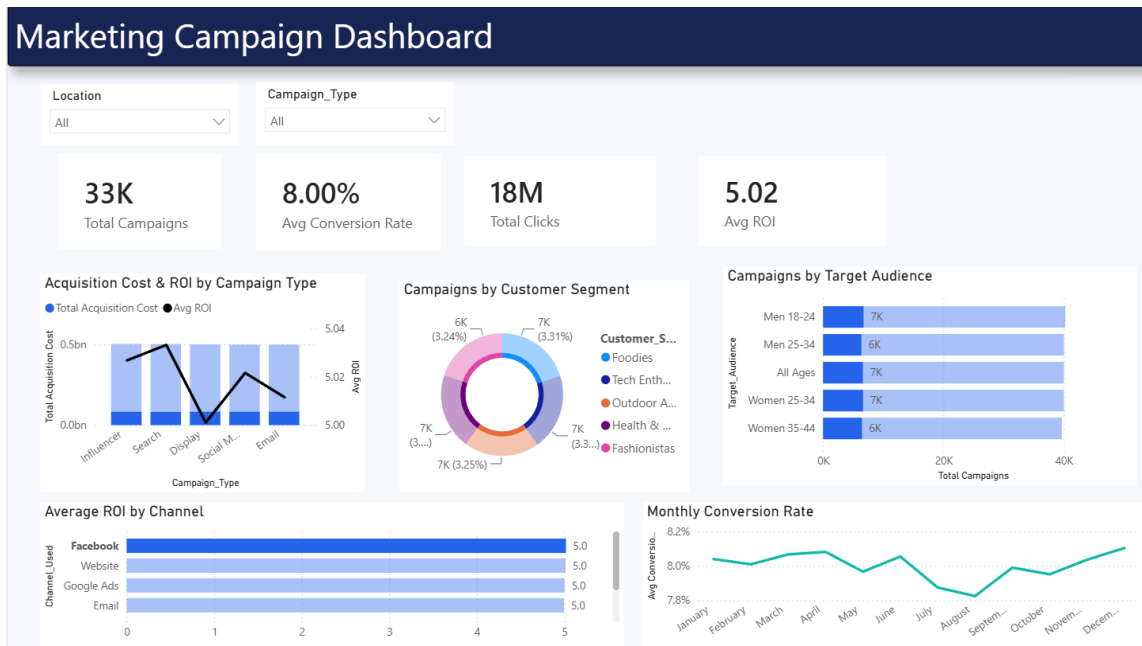


Figure 2: Marketing Campaign Dashboard – Filtered by Facebook Channel

Image 3rd:

In the dashboard, after selecting October on the Monthly Conversion Rate chart:

- All visuals are cross-filtered to show data for the month of October only.
- The KPI cards update to Total Campaigns = 17K, Avg Conversion Rate = 7.98%, Total Clicks = 9M, and Avg ROI = 5.02.
- Although conversion dips slightly below the yearly average in October, Avg ROI actually edges up to 5.02, showing that a lower conversion rate did not translate into weaker returns that month.
- This interaction helps management isolate month-specific performance and check whether short-term dips in conversion rate are affecting profitability.

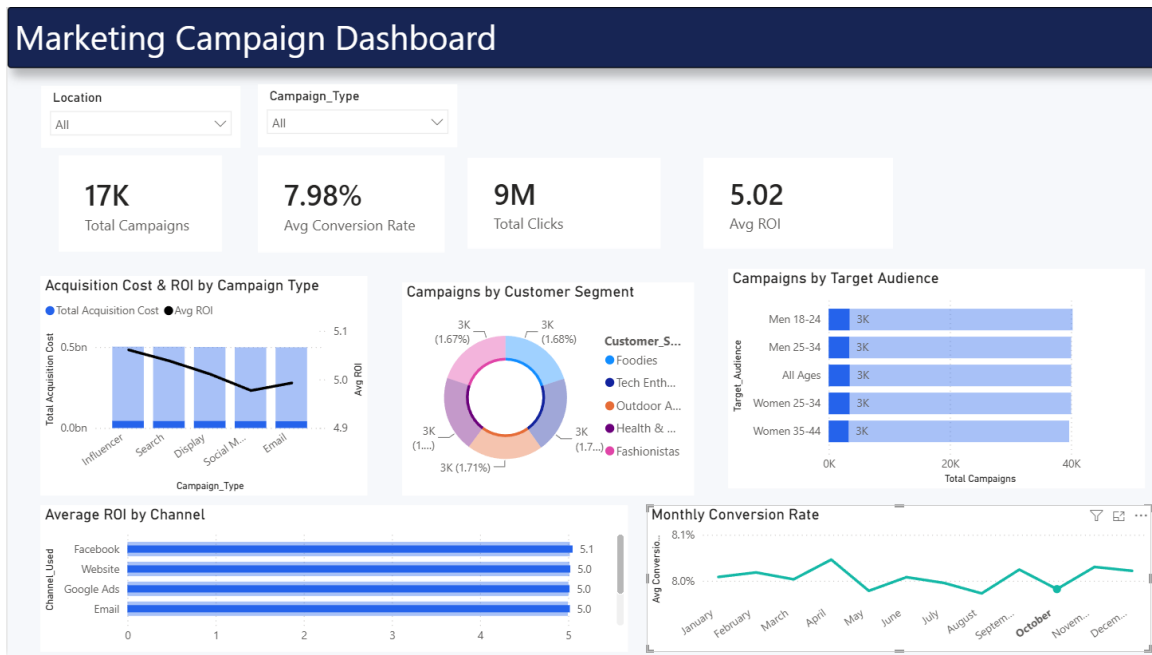


Figure 3: Marketing Campaign Dashboard – Filtered by October

DAX Measures Used

The following DAX measures were created on the marketing_campaign_dataset table to power the KPI cards and visuals in the dashboard:

1. Total Campaigns

```
Total Campaigns =
DISTINCTCOUNT('marketing_campaign_dataset'[Campaign_ID])
```

Calculates the count of unique campaigns in the dataset. Used to display the Total Campaigns KPI (200K).

2. Avg Conversion Rate

```
Avg Conversion Rate =
AVERAGE('marketing_campaign_dataset'[Conversion_Rate])
```

Calculates the average conversion rate across all campaigns. Used to display the Avg Conversion Rate KPI (8.01%).

3. Total Acquisition Cost

```
Total Acquisition Cost =
SUM(marketing_campaign_dataset[Acquisition_Cost])
```

Calculates the sum of acquisition cost spent across all campaigns. Used to power the Acquisition Cost & ROI by Campaign Type chart.

4. DateTable

```
DateTable =
CALENDAR(
    MIN(marketing_campaign_dataset[Date]),
    MAX(marketing_campaign_dataset[Date])
)
```

Creates a continuous date table spanning the full range of campaign dates. Used to support the Monthly Conversion Rate trend line and enable proper time-based filtering.

Business Insights

- The portfolio comprises 200K campaigns that together generated 110M clicks, with a stable Avg Conversion Rate of 8.01% and Avg ROI of 5.00 — indicating a mature, well-balanced campaign mix rather than one skewed by a few outlier campaigns.
- Acquisition cost is spread almost equally across the five campaign types (~₹0.50bn each), but Influencer and Search campaigns deliver the highest average ROI (5.01), while Social Media lags slightly (4.99) — suggesting a modest reallocation from Social Media toward Influencer and Search could lift blended ROI.
- Customer segments and target audiences are both very evenly represented, showing the dataset spans a genuinely diverse customer base rather than concentrating on one demographic.
- Facebook campaigns modestly outperform the overall base, with Avg ROI rising to 5.02 (vs 5.00 overall) when the channel is isolated, making it a candidate for incremental budget.
- October shows a slightly below-average conversion rate (7.98% vs 8.01% overall) but the highest Avg ROI observed in the drill-downs (5.02), reinforcing that conversion rate and ROI don't always move together and both metrics should be tracked jointly.
- Interactive slicers for Location and Campaign_Type allow management to drill down into specific combinations to identify the strongest-performing campaign segments and channels.

Conclusion

The Marketing Campaign Performance dataset was successfully imported into Microsoft Power BI and transformed into an interactive Marketing Campaign dashboard using KPI cards, a combo chart, a donut chart, bar charts, and a line chart. The dashboard effectively tracks campaign reach, conversion performance, acquisition cost, and ROI across campaign types, customer segments, target audiences, and channels. Overall, the interactive dashboard supports data-driven marketing planning and budget-allocation decisions by enabling management to monitor and evaluate campaign performance efficiently.